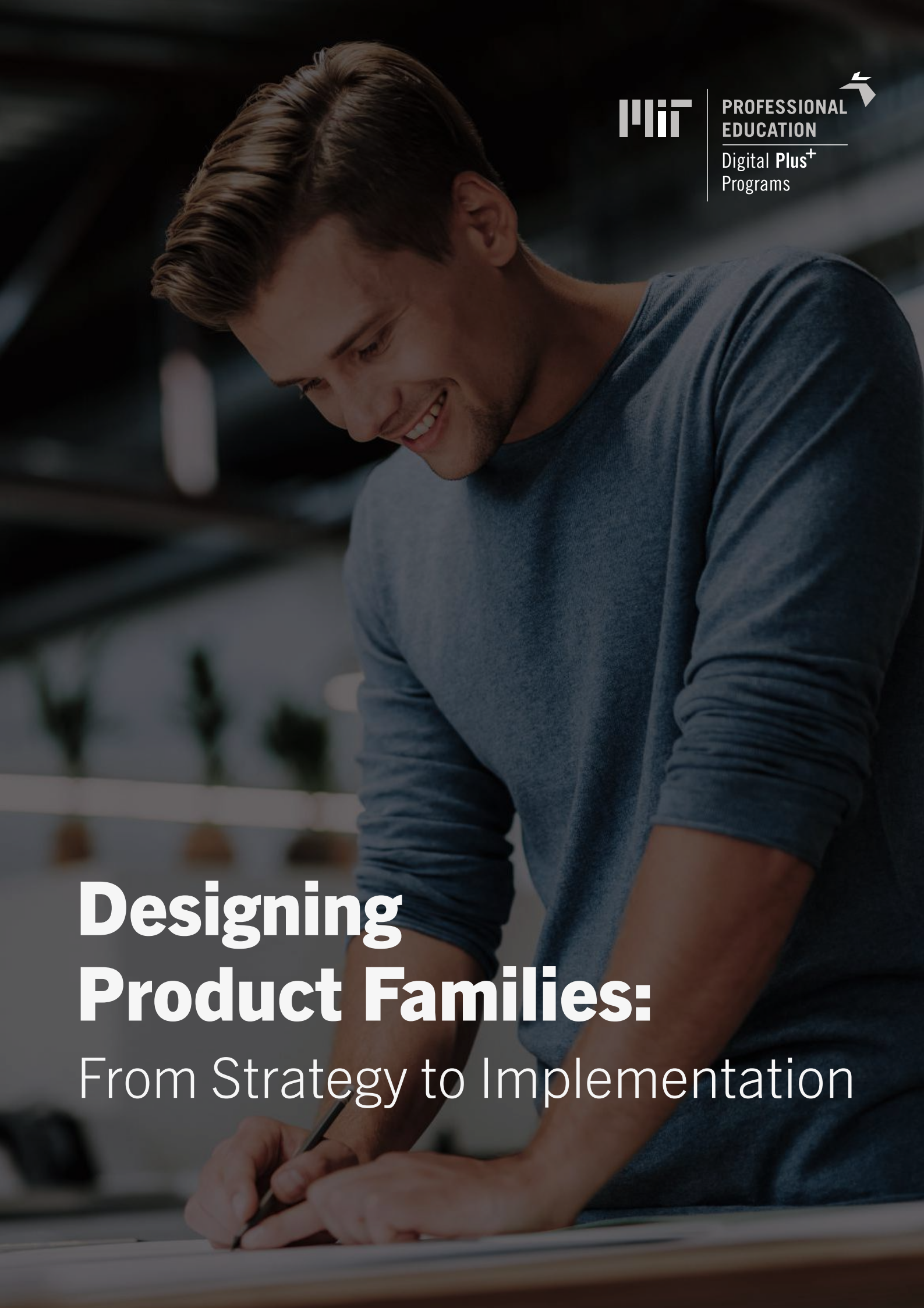
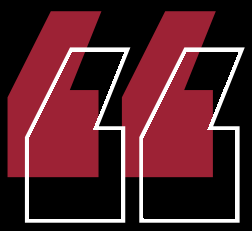


PROFESSIONAL EDUCATION
Digital Plus+ Programs

Designing Product Families:

From Strategy to Implementation



Designing platforms and product families combines the complexity of the market dynamic, strategy, R&D projects, and manufacturing in order to achieve long term success and the survival of companies

Prof. Olivier de Weck

Index

03 Personalization: The Manufacturing Sector's New Mantra

04 Benefits of Improving your Digital Strategy

05 About this Course

06 Key Takeaways

07 Participant Profile

08 Instructors

11 Course Outline

14 Certificate of Completion

16 About MIT Professional Education

17 Corporate Courses

18 The Beyond Online Methodology

19 Benefits of Joining the Alumni Community

Personalization: The Manufacturing Sector's New Mantra



Today's highly competitive marketplace is changing the way organizations do business. **The speed of innovation and mass customization offer new competitive advantages.**

In response, **organizations must implement strategies to design and develop product families** on a common platform, thereby meeting a wide variety of customer requirements and harnessing the power of economies of scale.

Your organization's growth in today's competitive marketplace will depend heavily on a strategy that optimizes **all stages of product creation, from architecture, implementation and management**.

Benefits of Improving your Digital Strategy

94%

of Fortune 1000 companies reported supply chain interruptions in 2021.

Source: Fortune

30%

was droppped in product development time due to the optimization of product families.

Source: NCDPD

20%

of costs can be saved by having an efficient product family or platform.

Source: PwC

About the Course



Over the course of 8 weeks, you will learn the keys to platform design with a hands-on approach to successfully increase the variety of a product line, shorten manufacturing lead times and reduce associated overhead costs. Thanks to MIT's signature hands-on methodology, you will combine a theoretical foundation with practical learning that will help you master current methods and tools in product family and platform design.

The key to success in manufacturing operations lies in the ability of the design and product families to adapt to the global market. In this regard, it is necessary to achieve short-term goals and maintain mass production efficiency standards.

8 weeks | 8-10 hours a week | Online

Key Takeways

Throughout this course, participants will:

01 Learn about the evolution of the industry, **from small-scale artisanal craftmanship to mass personalization.**

02 Understand fundamental principles of product creation, **such as manufacturing decisions, composition, and the design evolution process.**

03 List **parameters to quantify what is common within a product family.**

04 Identify the **main methods and tools used in the design of product families and platforms.**

05 Understand the utility of **comparing designs among different product platforms and families is useful.**

06 Learn about organizational issues that can arise, such as **divergence and platform management.**

07 Use platforms to **identify new market opportunities, increasing revenues** as a result



Participant Profile

This course is aimed at professionals looking to create orposition their products, services, and systems to innovate and compete in the market.

Research and development **managers.**

Platform **architects.**

Product and marketing **managers.**

Product line **leaders.**

Public administration and industry systems engineering **professionals.**

Profesionales de ingeniería de sistemas en la industria o la administración pública.

This program is for those who want to implement normalization, personalization, and platform use strategies to maximize cost savings and to increase their organization's capacity to offer a variety of personalized products and services.

It is helpful, but not necessary, to have basic training in engineering, as well as to have experience in commerce and accounting.

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Instructors



Prof. Olivier de Weck

Professor, Aeronautics and Astronautics and Engineering Systems, MIT

Olivier De Weck is a leader in systems engineering research. He focuses on how complex man-made systems such as aircraft, spacecraft, automobiles, printers, and critical infrastructures are designed and how they evolve over time. In addition, he studies strategic properties that have the potential to maximize lifecycle value and reduce programmed obsolescence.

Since 2001, his group has developed novel quantitative methods and tools that explicitly consider manufacturability, flexibility, commonality, and sustainability, among other characteristics. In his research, de Weck emphasizes excellence, innovation, and combining theory and practice.

> "Technology is the deliberate creation and use of objects and processes to solve specific problems"

De Weck holds a degree in industrial engineering from ETH Zurich and an M.S. in aeronautics and astronautics from MIT

He earned his Ph.D. in aerospace systems from MIT.

Previously, he was an engineering program manager on the F/A18 aircraft program at McDonnell Douglas.



Dr. Bruce Cameron

Director del MIT System Architecture Group

Dr. Bruce Cameron is director of the System Architecture Lab at MIT.

His areas of research include technology strategy, system architecture and product platform management.

He is currently the Faculty Director of MIT´s online certificate “Architecture and System Engineering” and is working on various research projects: Analyzing the costs of supplier replacement and retention incentives in driven vehicles for hire (ride hailing), the effects of increased autonomy in commercial trucking on fleet management, and the effects of one-pilot operations on commercial flights.

> “The best companies in the world use platforms to compete. The key to this strategy is understanding how to tell the difference between consumer choice and reinventing the wheel.”

Dr. Cameron has led research projects for Amazon, Chevron, Sikorsky, Nokia, Caterpillar, and NASA.

His work in Sloan Executive Education has been rated as the highest level of Executive Education program at MIT for several years

He is a co-founder of Technology Strategy Partners and was the consultant for twelve companies in initial phases of online commerce platforms.



Prof. Timothy W. Simpson

Lecturer en MIT Professional Education

Timothy Simpson is a Professor of Mechanical Engineering and Industrial Engineering at the Pennsylvania State University. His research interests include product family design and platform-based product development, multidisciplinary design optimization, trade space exploration, and additive manufacturing.

He has over 250 publications in peer-reviewed journals and conference proceedings. Simpson is the lead editor on two books.

His research has been supported by a variety of federal and state agencies, including NSF, ONR, DARPA, EDA, and DCED, as well as numerous companies, ranging from startups to multinational corporations. He has worked with over 30 different companies to apply his research methods to improve their product lines and product development practices. He helped establish and manage the Product Platforms Group on LinkedIn and is the only faculty member who has been honored by the Penn State Engineering Alumni Society, both for his teaching and his research.

> "Courses often don't ask the question, 'How can we really design and take advantage of these processes?' and that question drives my courses."

Professor Simpson was a visiting researcher at the Multidisciplinary Optimization Branch of the NASA Langley Research Center in Hampton, VA

He was involved in the Applied Mathematics and Statistics Group at The Boeing Company in Seattle, WA

He has received awards such as the ASME Ben C. Sparks Medal (2014), the ASEE Fred Merryfield Award (2011), and the Penn State President's Award for Excellence in Academic Integration (2007)

Course Outline

How do we conceptualize Platforms at MIT Professional Education?
For the Faculty of this program, a platform is the productive space of a family of products in which all these products or services share a series of common components, technologies and know-how applied in order to save production costs, optimize the supply chain and increase company profits by sharing common elements.

Module 1
Fundamental Principles of Platform-Based Manufacturing
In this module, we will look at some of the conceptual foundations on which platform systems are understood. We will have the opportunity to learn about the ways in which different companies have established a conception of design and industrial production based on platforms and exploitation methodologies. We will approach some case studies of real companies and products. After this, we will analyze the paradoxes and contradictions that arise in this field. Although platforms offer many opportunities, going through the different phases prior to their creation process involves several difficulties.

1. Mindset
2. Examples and definitions
3. Approaches
4. Use and paradoxes

Module 2
The advantages of platforms and implementation strategy
In this module, we will look at the benefits and advantages of product platforms, as well as the strategy behind them. We will analyze the results of case studies and determine what the benefits of the platforms are. We will also differentiate between product lines and product platforms. Finally, we will study the eight key decisions that define a company's platform policy or strategy.

1. Competitive advantage
2. Product line strategy
3. Advantages of platforms
4. Profits and investments predictions
5. Platform Strategies

Module 3
Product and Platform Architecture
In this module, we will analyze the architecture of products. First, we will know what architecture is and we will look at different exchange strategies. Next, you'll learn how to prioritize architectural decisions based on connectivity and responsiveness. Finally, you will learn how to create a Design Structure Matrix (DSM) and identify modularity in the product architecture.

1. Architecture definition
2. Architectural decisions
3. Form and Function
4. Breakdown and Design Structure Matrix (DSM)
5. Modularity and interfaces

Module 4
Platform Analysis
In this module, you will learn methods and tools for product platform analysis: maps and metrics. You will learn tools to analyze and measure the effectiveness of a product platform including aggregate project planning, commonality metrics for platform management, and how commonality enables production deferral and manufacturing.

1. Product families schemes
2. Aggregate project planning
3. Sharing strategies
4. Indices of communality
5. Metrics

Module 5
Platform Design
In this module, you will learn how to assess variety using a Generational Variety Index (GVI). You will use an integrated approach to product family redesign that benchmarks commonality against desired variety at the component level and assesses the impact of any proposed changes. You will also use an optimization framework and determine the extent of the platform.

1. Architecture overview
2. Determine differentiation
3. Product platform market comparison
4. Product family redesign and optimization

Module 6
Platform Planning
During this module, you will reinforce your understanding of overdesign requirements and follow a step-by-step process to build flexibility into a platform. You will also learn how to apply the platforms within software and services.

1. Planning and flexibility
2. Flexible platform design process
3. Software platforms
4. Non-traditional applications

Module 7
Industrial Platforms and Bilateral Markets
This week, you will study platform management strategies. We'll use examples from the MIT Commonality Study to examine both common successes and failures to discover where companies fail. You will learn the importance of being aware of communality and decision rights (and who should have them). You will also see how contracts and investments interact with the commonality and financial benefits of a platform.

1. Common vision and organizational skills
2. Divergence and management levers
3. Contracts and roles

Module 8
Two-sided markets and industrial platforms
In this last module, we will examine the relationship between platforms and double-sided markets. First, we will define double-sided markets to understand how platforms enable them. You will learn to quantify double-sided markets with metrics to understand their evolution. Then, we will see a detailed case study on mobile phones and modular accessories.

1. Twp-sided markets
2. Mathematical model
3. Market validation

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Certificate of Completion

All participants who successfully complete the course will receive an official certificate from MIT Professional Education and 6.4 MIT CEUs*

In order to obtain MIT CEUs, participants must complete a required MIT CEU confirmation form. MIT CEUs are calculated based on the number of learning hours in each course.



*The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



Welcome to the gateway

to MIT knowledge and expertise for professionals around the world.

155

Participants from 155 countries

+1600

More than 1,600 online activities

92%

92% of Professional Alumni rated their experience as outstanding

About MIT Professional Education

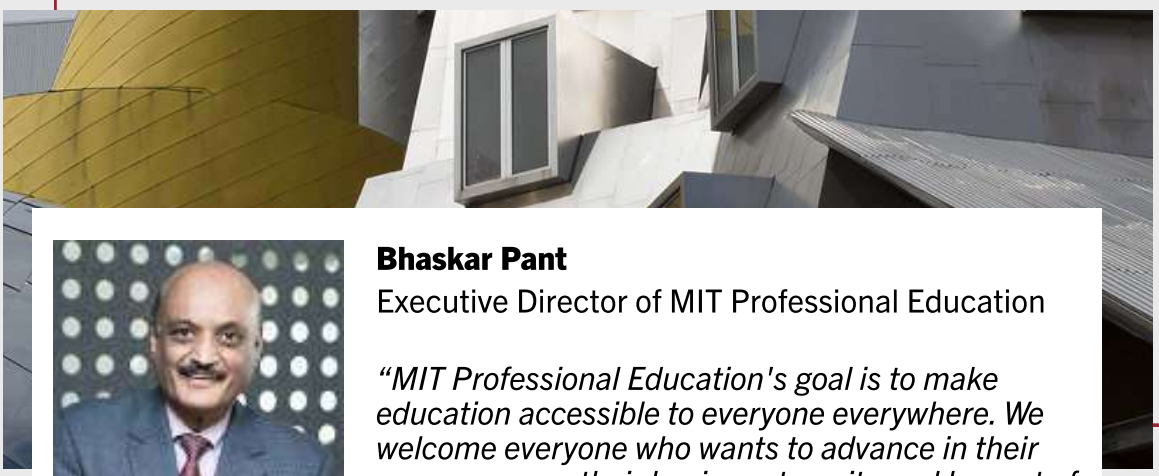
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind.**





Bhaskar Pant
Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:
Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs:
Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:
Available in Open Enrollment courses with a minimum of participants.

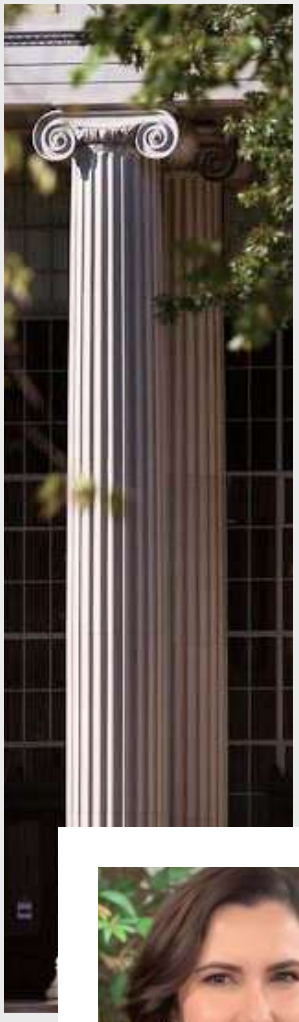


For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

The Beyond Online Methodology

MIT Professional Education is **revolutionizing the digital learning experience**. In fact, we are no longer talking about online-only, but rather an **interactive and collaborative learning experience** that's digitally powered: a **Beyond Online experience.**

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology.You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.



Clara Piloto
Director of Global Programs at MIT Professional Education

"Personalization of products is the new slogan of the manufacturing industry. The method in which companies are doing business is undergoing rapid change. Designing product families with innovation and mass customization in mind offers an emerging competitive advantage."

Benefits of Joining the Alumni Community



MIT Professional Education offers **a number of benefits for participants** who successfully complete our courses:

1. **15% discount** for MIT Professional Education online courses and short-duration, in-person programs.
2. **Continued access to the virtual campus for six months after completion** of the course.
3. **Exclusive announcements** of new courses, programs, and events through our **virtual Alumni Campus: Professional Network.**
4. Invitation to the **MIT Professional Education Alumni LinkedIn group.**
5. **Networking opportunities** with other MIT Professional Education Alumni.

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PROFESSIONAL EDUCATION
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Are you ready to become a leader in the latest technologies?

Frequently Asked Questions

global alumni
-excellence or nothing-

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